

News Momentum and Price Momentum

Junwei Yang

November 17, 2025

Abstract

This study examines the relationship between price momentum and news narratives in commodity futures markets. By extracting price drivers from financial news, we developed a novel "theme score" to create a news-enhanced momentum strategy. An empirical test on crude oil futures shows our news-enhanced momentum strategy outperforms traditional momentum and buy-and-hold approaches, yielding an annualized Sharpe ratio of 0.97 between January 2023 and May 2025. Our findings confirm the efficacy of incorporating news narratives in improving trading performance.

Keywords: News narratives, momentum strategies, commodity futures, crude oil

JEL Classification: G11, G12, G14, G17

The human brain has always been highly tuned toward narratives, whether factual or not, to justify ongoing actions, even such basic actions as spending and investing.

—Robert Shiller ([2017](#))

1 Introduction

Financial markets often coalesce around salient narratives—or *themes*—, which usually concentrate investors' attention. For instance, in mid-June 2025, geopolitical risk dominated the crude oil market: on June 13, reports of an Israeli military strike on Iran prompted fears of supply disruption and a 7% surge in the July WTI contract; volatility persisted until June 24, when a tentative

Israel–Iran ceasefire eased regional risk and the August WTI contract fell by 6%. Likewise, the 2024 cocoa rally was largely narrated through persistent supply concerns linked to sustained heat in West Africa.

This paper formalizes the link between news narratives and returns by positing that, at each point in time, market returns are generated under the influence of one or more latent themes. We operationalize this idea by extracting theme-specific signals from financial news and mapping them to prospective price movements. Bolstered by the literature on gradual information diffusion and limited attention—as well as behavioral biases that can induce persistence in price responses (Daniel et al., 1998; Hong and Stein, 1999; Goyal et al., 2025)—we hypothesize that news-driven themes can exert persistent, measurable effects on returns.

Our empirical analyses use news and price histories for crude oil futures. We develop a return decomposition that attributes realized returns to contemporaneous themes and analyze the dynamics of cross-theme attention. We also propose a mixture model of return generation, in which the theme score vector acts as a dynamic decision signal. We assess the model’s statistical properties and trading performance relative to canonical benchmarks, including time-series momentum.

A large literature integrates non-price information into asset pricing and forecasting using textual sources such as corporate disclosures, earnings calls, social media, and financial news, enabled by advances in document retrieval and NLP (Loughran and McDonald, 2016; Gentzkow et al., 2019; Xing et al., 2018; Du et al., 2025). Representation learning spans dictionary-based counts, topic models, and modern embeddings, facilitating applications from sentiment and tone to readability, uncertainty, and news-implied volatility (Antweiler and Frank, 2004; Tetlock, 2007; Bollen et al., 2011; Jegadeesh and Wu, 2013; Ke et al., 2019; Li, 2008; Manela and Moreira, 2017; Baker et al., 2016). Within asset-pricing factors, commodity returns have been linked to storage, convenience yield, and supply–demand shocks, as well as cross-sectional and time-series momentum (Fama and French, 1987; Gorton and Rouwenhorst, 2006; Delle Chiaie et al., 2022; Bessembinder et al., 1995; Miffre and Rallis, 2007; Moskowitz et al., 2012). Theme discovery in text has been explored via unsupervised topic models and attention-based constructs (Doyle and Elkan, 2009; Bybee et al., 2024), and recent work investigates thematic allocations in portfolio contexts (Somefun et al., 2023).

On the feature side, text-based signals are typically represented as sen-

timent scores, generic word sets, or numerical embeddings, all of which are difficult to map onto economically interpretable commodity drivers. On the modeling side, the link between narrative attention and return persistence remains under-specified. Conventional momentum is defined on and inferred solely from prices rather than from the evolution of theme-specific news attention, leaving the dynamics of attention effectively under-identified.

We address these gaps by (i) defining a pre-specified, economically grounded taxonomy of commodity themes and using classification with a language model to map each piece of news into those themes, yielding interpretable and empirically predictive *market theme scores*; (ii) constructing a return decomposition and a mixture model of return generation in which the theme score vector acts as a dynamic decision signal, directly linking the evolution of narrative attention to expected returns. We integrate theme scores into an allocation framework that complements and improves on time-series momentum in commodity futures. Relative to unsupervised topic models (Doyle and Elkan, 2009; Bybee et al., 2024), our approach trades discovery for interpretability and stability, aligning signals with identifiable supply, demand, currency and geopolitical mechanisms in commodity markets.

The remainder of the paper proceeds as follows. Section 2 describes the news and price datasets. Section 3 develops the conceptual framework, the classification pipeline, and the construction of theme scores. Section 4 presents the mixture model of returns and discusses its identification and estimation. Section 5 reports empirical results and trading performance in crude oil relative to standard alternatives. Section 6 concludes.

2 Data

We utilize a unique dataset comprising daily commodity futures prices and corresponding financial news from October 2021 to May 2025.

2.1 Price History

We collect daily price history for continuous front-month futures contracts of Crude Oil WTI (CLc1) over the observation period from LSEG (former Refinitiv). This dataset includes the exchange date, closing price, percentage change, open interest, and volume. Among those, we use exchange date and return(percentage change) fields. Figure 1 presents the daily return history

and its associated distribution for WTI crude oil futures. We observe volatility clustering effect and a close relationship between return and events. We label influential event triggering extreme positive or negative returns in the Figure.

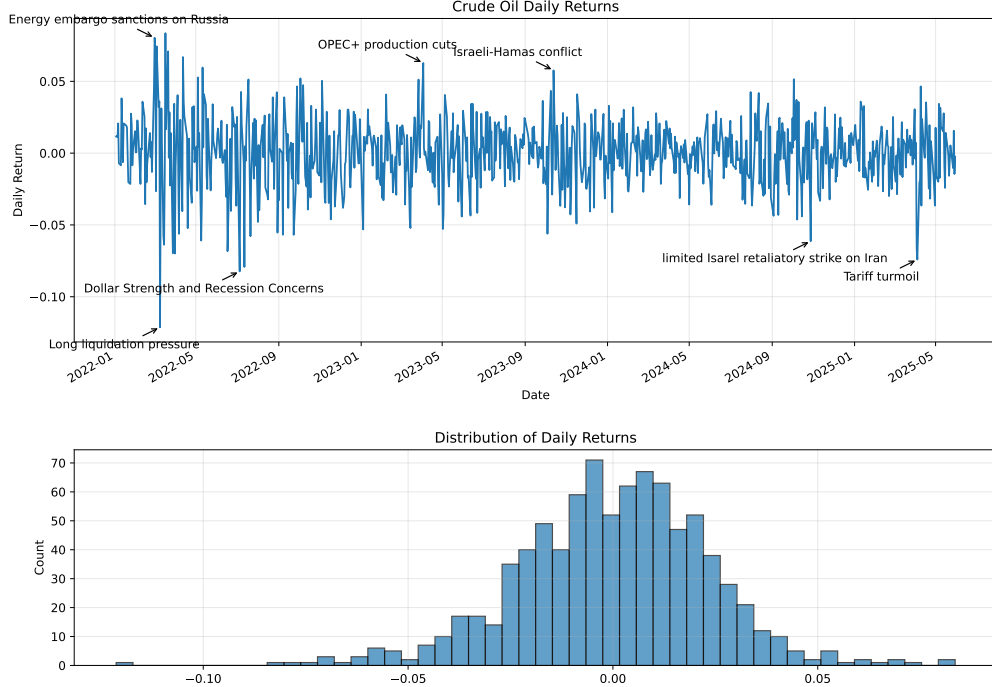


Figure 1: WTI Daily Returns. The upper subplot reports daily return time series of WTI Crude Oil continuous future(Jan 2022-May 2025) and the lower subplot reports the distribution of WTI daily returns during the same period.

2.2 News

Our news data is collected from Barchart news articles¹. For each calendar date (represented in YYYY-MM-DD format) in Central Daylight Time (CDT, UTC-5), we build a corresponding 24-hour timestamp range in Coordinated Universal Time (UTC). This standardized UTC timestamp range is then used to query and retrieve all news articles published on that specific day through Barchart API. Rather than focusing on single events, Barchart news articles typically provide concise summaries that document price movements of various commodities, accompanied by inferred explanations or contextual factors. Representative headlines include examples such as “Crude Oil Rallies Ahead of Friday’s Trump-Putin Summit” and “Dollar Sees Support from Strong US PPI Report.” The raw data for each article includes a unique news ID, publication

¹<https://www.barchart.com/news/commodities>

time, title, URL, and a short excerpt.

Subsequently, we employ a language model (deepseek-reasoner) to process raw information and format a structured result, summarizing key price drivers for each commodity from news sources. These identified drivers are then utilized as input for a secondary classification language model (Gemini-2.5-pro), which categorizes them according to a predefined set of themes. An example of the resulting structured data format is provided in Table 1. This table presents the daily price drivers, with their corresponding identified themes indicated in parentheses. Through this process, we establish a mapping between daily market returns and their associated, thematically categorized price drivers.

Table 1: News Format.

Date	Commodity	Key Drivers
2023-01-03	Crude Oil	Dollar rally (currency); Stock market declines (markets)
2023-01-04	Crude Oil	Chinese demand concerns (demand)
2023-01-05	Crude Oil	Chinese fuel demand (demand)
2023-01-06	Crude Oil	Weak dollar (currency); Equities rally (markets); Strong U.S. stocks (markets)
2023-01-09	Crude Oil	Dollar index slump (currency); Chinese fuel demand (demand)
2023-01-10	Crude Oil	China demand recovery (demand)

Note: Key crude oil price drivers are extracted by language models from news article titles and short excerpts published on the corresponding date. The thematic category for each driver is indicated in parentheses.

The prompts input into the language model encompass five themes:

- Supply: related to production, logistics, and producer (OPEC+) decisions.
- Demand: Factors related to consumption, demand forecasts, and macroeconomic indicators (GDP, manufacturing data, recession fears)
- Geopolitics: Factors from political events, sanctions, or regional conflicts.
- Currency: Factors related to foreign exchange rates, primarily the U.S. Dollar.

- Markets: Factors related to financial market sentiment, equities, investor risk appetite, financial sector stability, and technical trading.

The first four themes: supply, demand, geopolitics, and currency are fundamentally linked to crude oil's dynamics and dominate crude oil related news narratives, while the market theme serves as a residual category for speculation or noise trading behavior when the market movement can't be attributed to fundamental reasons easily. Because we focus on the integration of news information, further investigation focuses on the first four themes.

3 Return Decomposition and Theme Score

In this section, we introduce key concepts used to formulate a return decomposition model, and illustrate how we calculate theme scores, which we use to measure market attention across themes.

3.1 Return Decomposition to Activated Theme

Definition 1 (K-theme Market). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We define a K-theme Financial Market as a stochastic process represented by the tuple $(\Delta_t, D_t)_{t=1,2,\dots,T}$. Here $\Delta_t \in \mathbb{R}$ is the market return at time t and D_t is the activation set of the market at time t . The integer K denotes the total number of themes. The information structure is described by the filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$, $\mathcal{F}_t = \sigma(\{(\Delta_s, D_s) \mid 1 \leq s \leq t\})$, including the history of both market returns and theme activations.*

If theme i in the activation set D_t , we call theme i is active and the market is under theme i at time t . We posit that when a theme is active, its underlying dynamics contribute to the market's price movement. The observed market return at time t , Δ_t , is therefore a composite result of all themes that are active at that time.

3.2 Theme Score

The foundational assumption of our model is that price movements are driven by new information that deviates from the market consensus. This "market consensus" refers to the information contained within the price filtration, $\mathbb{G} = (\mathcal{G}_t)_{t=1,2,\dots,T}$, $\mathcal{G}_t = \sigma(\Delta_1, \dots, \Delta_t)$. Consequently, to separate the information increment from consensus, we define a variable $\Delta_{i,t} \in \mathbb{R}$, which we term *theme-specific information increment*. This variable represents the new information

arriving at time t that is pertinent to theme i and is not captured by the historical price series alone. That is, $\Delta_{i,t}$ is adapted to the full information filtration $\mathbb{F}$ (which includes news information) but not to the price filtration $\mathbb{G}$. Then the theme-specific increment is used to calculate theme score $S_{i,t}$. We first note the relationship between theme score $S_{i,t}$ and future return.

Definition 2 (Theme Score). *We define the Theme Score, $S_{i,t}$, as a continuous hidden state variable representing the latent prominence of theme i at time t . Let $Z_t \in \{1, \dots, K\}$ be a latent random variable with a categorical distribution of K themes, such that:*

$$P(Z_t = i) = \pi_{i,t}, \quad \text{for } i = 1, \dots, K \quad \text{and} \quad \sum_{i=1}^K \pi_{i,t} = 1.$$

The random variable Δ_t is then distributed as follows, conditional on the value of Z_t :

$$\Delta_t | (Z_t = i) \sim S_{i,t-1} + \epsilon_{i,t}.$$

The notation $A \sim B$ means that the random variable A has the same probability distribution as the random variable B . $\epsilon_{i,t}$ is used to describe the distribution, has a zero mean, and is independent of $S_{i,t-1}$.

With Definition 2, the market return is determined by their corresponding scores from the previous period, $S_{i,t-1}$.

We model that theme score $S_{i,t}$ is influenced by immediate impact from $\Delta_{i,t}$ and lingering effects from past increments, expressed in a moving average form:

$$S_{i,t} = \sum_{j=0}^{p-1} a_j \Delta_{i,t-j}, \tag{1}$$

where the coefficients $\mathbf{a} = (a_0, \dots, a_{p-1})$ are the weights of the filter over a window of size p . Equation (1) builds the relationship between aggregated theme score and daily information increment.

Example 3.1 (Relation to Autoregressive Models). *When there's only one theme ($K = 1$), if we assume $\Delta_{i,t} = \Delta_t$. Substituting these into our model, we have in probability*

$$\Delta_t = S_{t-1} + \varepsilon_t,$$

then plug in Equation (1):

$$\Delta_t = \sum_{j=0}^{p-1} a_j \Delta_{t-j-1} + \varepsilon_t,$$

which is the form of an $AR(p)$ model.

This is clear because if there's only one scheme, the labeling doesn't provide any additional information therefore all the information is contained in price history. We will revisit the close relationship of our model with multiple themes to AR model's correlation structure later.

We give an illustration of our computational flow in Figure 2, where an example of a four-theme market with a filter window of size 4 is displayed. The next return's theme category and its value is sampled given the information of previous theme scores.

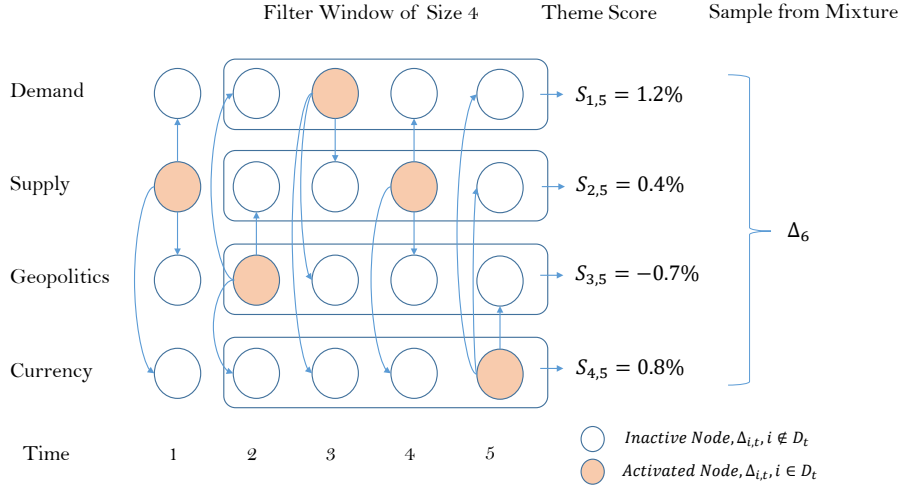


Figure 2: Computational Flow. This diagram illustrates our model's computational process for a four-theme market (Demand, Supply, Geopolitics and Currency). A filter window of size 4 processes the theme-specific information increment time-series. Activated theme is colored in orange and inactive theme is in white. Arrow lines denote the impact from activated theme to inactive themes. A theme score is calculated for each theme after the filtering process. The next return Δ_6 is sampled from a mixture model of previous theme scores $S_{i,5}$.

4 A Theme-Based Model for Price Movement

Building upon the concepts presented in Section 3, in this section, we propose a data generating process that mimics financial time series returns by integrating components driven by news themes.

4.1 Model Setting

We assume that the return Δ_t follows a mixture Gaussian distribution, characterized by the probability density function:

$$p_{\Delta_t}(x) = \sum_{i=1}^K \pi_{i,t} \mathcal{N}(x | \mu_{i,t}, \sigma_{i,t}^2), \quad \sum_{i=1}^K \pi_{i,t} = 1, \quad \pi_{i,t} \geq 0, \quad (2)$$

where K is the total number of themes.

We assume that the volatility is constant across all themes and over time, denoted by σ :

$$\sigma_{i,t} = \sigma.$$

We then focus on the evolution scheme of mean $\mu_{i,t}$ and sampling probability $\pi_{i,t}$.

Update Equation for the Mean. To distinguish the impact of intrinsic price dynamics from the information increments from news labels, we introduce a consensus operator C designed to extract information from price movements.

Definition 3 (Consensus Operator). *A consensus operator $C : \mathbb{Z} \rightarrow \mathbb{R}^n$ is a function that maps a time t to an n -dimensional vector, which represent a summary of information from the historical price return series $\{\Delta_s\}_{s < t}$.*

We illustrate with two choices of the consensus:

- **Autoregressive Model:** An AR(p) structure, $C(t) = \Phi(B)(\Delta_t) = \sum_{k=1}^p \phi_k \Delta_{t-k}$, where the consensus is a weighted average of past returns.
- **Martingale Model:** $C(t) = E[\Delta_t | \mathcal{F}_{t-1}]$, where the consensus is the conditional mean of the process given the information up to time $t-1$.

The difference between the current return and its consensus is the information increment where news labeling might contribute:

$$\tilde{\Delta}_t = \Delta_t - C(t). \quad (3)$$

Furthermore, we assume that the information increment $\tilde{\Delta}_t$ impacts both activated and inactive themes, but in different ways. Let D_t be the set of activated themes at time t . The theme-specific increment, $\Delta_{i,t}$, is defined as:

$$\Delta_{i,t} = \tilde{\Delta}_t \mathbb{I}\{i \in D_t\} - \beta \tilde{\Delta}_t \mathbb{I}\{i \notin D_t\}, \quad (4)$$

where the parameter β governs the permeation of information from active to inactive themes. When $\beta > 0$, a positive shock to an active theme creates a negative pressure on inactive themes, introducing a mean-reverting mechanism into the market.

Finally, the theme score $S_{i,t}$ is a weighted sum of past theme-specific information increments shown in Equation (1).

The theme- i mean $\mu_{i,t}$ is dictated by its score $S_{i,t-1}$,

$$\mu_{i,t} = E[S_{i,t-1} + \epsilon_{i,t} \mid \mathcal{F}_{t-1}] = S_{i,t-1}. \quad (5)$$

This setting correlates a theme's impact with the expectation of future returns.

Update Equation for Sampling Probability. The second core component of the model is the evolution of the theme activation probabilities, $\pi_{i,t}$. We use a parameter we term probability-mixture c to control the extent of self-reinforcing patterns in activation. A small c indicates that the activation of a theme in the next period depends much on its current theme score, while a value of c close to 1 indicates an almost equally likely chance for all themes to be activated in the next period.

$$\pi_{i,t} = (1 - c) \frac{|S_{i,t-1}|}{\sum_{k=1}^K |S_{k,t-1}|} + \frac{c}{K}, \quad c \in [0, 1], \quad (6)$$

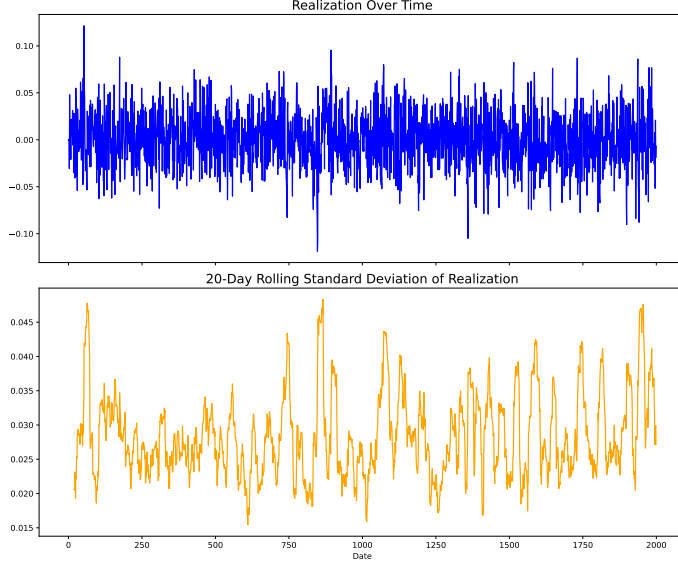
where K is the total number of themes.

With this architecture, we claim that this model has the flexibility to generate several stylized facts of financial return series, including: volatility clustering effect, leptokurtic distribution, ideal autocorrelation structure and a balance between momentum effect and mean-reverting effect. The mixture design naturally generates higher volatility, while the interplay between the mean-correction and sampling parameters provides the flexibility to capture momentum and mean-reverting effects.

We generate a simulated trajectory of daily returns data under the setting: annualized volatility=0.4, ProbMixture=1.0, MeanReversionBeta=0.4, filter-

ing weights before unification= $(0.2, 0.7, 1.0)$, whose trajectory plot along with sample statistics are presented in Figure 3.

(a) Simulated Returns and 20-Day Rolling Volatility



(b) Sample Statistics

Moments	
Count	2000
Mean(%)	0.04
Min	-11.9%
Max	12.1%
Vol(Ann.)	0.467
Skew	-0.008
Kurt	0.236
Autocorrelations	
AC(1)	0.026
AC(2)	0.061
AC(3)	0.024

Figure 3: Simulated Returns. Panel(a) shows a time series plot of a simulated daily return trajectory generated by the proposed theme-based model, along with its 20-day rolling standard deviation in the lower subplot. Panel(b) presents sample statistics. Sample mean is shown in %, Vol(Ann.) denotes annualized volatility, Kurt is excess kurtosis, and AC(k) denotes autocorrelation with lag k.

4.2 Statistical Properties of a Simplified Model

In this section, we analyze the statistical properties of the model introduced in Section 4.1. We first derive analytical results under a simplified setting. Subsequently, we investigate the effects of various parameters under more complex scenarios through numerical simulations.

For analytical tractability, we adopt the following simplifying assumptions:

1. **Consensus:** The consensus is the expectation $C(t) = E(\Delta_t)$.
2. **Single activation:** Only one theme is activated at any given time.
3. **Uniform activation:** Each theme has an equal probability of being activated (i.e., $c = 1$).

Proposition 1 (Expectation). *Under assumption (1)-(3), $E(\Delta_t) = 0$.*

Proof. Under the uniform activation assumption,

$$E(\Delta_{t+1}) = E\left[\frac{1}{K} \sum_{i=1}^K S_{i,t}\right] = \frac{1}{K} E\left[\sum_{i=1}^K \sum_{j=0}^{p-1} a_j \Delta_{i,t-j}\right] \quad (7)$$

Because only one theme is activated at each time, adding all the $\Delta_{i,t-j}$ gives

$$\sum_{i=1}^K \Delta_{i,t} = (1 - (K-1)\beta) \tilde{\Delta}_t, \quad (8)$$

Plug Equation (8) back into Equation (7):

$$E(\Delta_{t+1}) = E\left[\frac{1}{K} \sum_{j=0}^{p-1} a_j (1 - (K-1)\beta) (\Delta_{t-j} - E[\Delta_{t-j}])\right] \quad (9)$$

$$= \frac{1}{K} \sum_{j=0}^{p-1} a_j (1 - (K-1)\beta) E[\Delta_{t-j} - E[\Delta_{t-j}]] = 0. \quad (10)$$

□

In addition, under the uniform activation assumption, for each theme score,

$$E(S_{i,t}) = \sum_{j=0}^{p-1} a_j E[\Delta_{i,t}] = \sum_{j=0}^{p-1} a_j E[\tilde{\Delta}_t (\mathbb{I}\{i \in D_t\} - \beta \mathbb{I}\{i \notin D_t\})],$$

by the symmetry from uniform activation assumption,

$$E(S_{i,t}) = \sum_{j=0}^{p-1} a_j E[\tilde{\Delta}_t] E[(\mathbb{I}\{i \in D_t\} - \beta \mathbb{I}\{i \notin D_t\})].$$

Since $E[\tilde{\Delta}_t] = E[\Delta_t - E[\Delta_t]] = 0$, therefore

$$E(S_{i,t}) = 0.$$

Then we derive the variance.

Proposition 2 (Variance). *The variance of the model can be expressed as $\sigma^2 + \tilde{\sigma}^2$, where σ is the volatility of each component theme, and $\tilde{\sigma}^2 := E[S_{i,t-1}^2]$, if $\{S_{i,t}\}$ is weakly stationary and square-integrable.*

Proof.

$$\begin{aligned}
\text{Var}(\Delta_t) &= E[(\Delta_t - E[\Delta_t])^2] = \frac{1}{K} \sum_{i=1}^K E[(\Delta_t - E[\Delta_t])^2 | i \in D_t] \\
&= \frac{1}{K} \sum_i E[(\Delta_t - \mu_{i,t} + \mu_{i,t} - E[\Delta_t])^2 | i \in D_t] \\
&= \frac{1}{K} \sum_i E[(\Delta_t - \mu_{i,t})^2 + (\mu_{i,t} - E[\Delta_t])^2 | i \in D_t]
\end{aligned}$$

Because $E[\Delta_t] = 0$ and $\mu_{i,t} = S_{i,t-1}$, therefore

$$\text{Var}(\Delta_t) = \frac{1}{K} (K\sigma^2 + \sum_{i=1}^K E[S_{i,t-1}^2]) \geq \sigma^2$$

□

Further, we investigate its autocorrelation structure to unveil the effect of mean correction parameter β . We assume the underlying process $\{\Delta_t\}$ is weakly stationary.

Proposition 3 (Autocorrelation). *The model exhibits an autocorrelation structure analogous to that of an AutoRegressive (AR) process. A filter window of size p in Equation (1) implies an AR(p) model for the return series $\{\Delta_t\}$. For the specific case when $p = 2$, the lag-1 autocorrelation is given by:*

$$\text{Corr}(\Delta_t, \Delta_{t-1}) = \frac{a_0}{\kappa - a_1},$$

where $\kappa = \frac{K}{1-(K-1)\beta}$. If $\beta = \frac{1}{K-1}$, it has a zero autocorrelation.

Proof. Under the assumption of stationarity, we can denote autocovariance at lag k as $\gamma(k) = \text{Cov}(\Delta_t, \Delta_{t-k})$. Because Δ_t has a zero mean, for $k = 1$,

$$\begin{aligned}
\gamma(1) &= \text{Cov}(\Delta_{t+1}, \Delta_t) = E[\Delta_{t+1}\Delta_t] \\
&= E\left[\left(\sum_{i=1}^K \frac{1}{K} S_{i,t}\right) \Delta_t\right] = E[S_{i,t}\Delta_t] \quad (\text{for any } i) \\
&= E\left[\left(\sum_{j=0}^{p-1} a_j \Delta_{t-j} (\mathbb{I}\{i \in D_{t-j}\} - \beta \mathbb{I}\{i \notin D_{t-j}\})\right) \Delta_t\right] \\
&= \sum_{j=0}^{p-1} a_j E[\Delta_t \Delta_{t-j} (\mathbb{I}\{i \in D_{t-j}\} - \beta \mathbb{I}\{i \notin D_{t-j}\})].
\end{aligned}$$

By the uniform activation assumption, and the same zero mean of $E[S_{i,t}]$, $E[\Delta_t \Delta_{t-j}]$ is independent of the initial category D_{t-j} , thus we could separate the two parts. The expectation of the indicator term is:

$$E[\mathbb{I}\{i \in D_{t-j}\} - \beta \mathbb{I}\{i \notin D_{t-j}\}] = \frac{1}{K} - \beta \frac{K-1}{K} = \frac{1 - (K-1)\beta}{K}.$$

Substituting this result into the expression for $\gamma(1)$ yields:

$$\gamma(1) = \frac{1 - (K-1)\beta}{K} \sum_{j=0}^{p-1} a_j E[\Delta_t \Delta_{t-j}] = \frac{1 - (K-1)\beta}{K} \sum_{j=0}^{p-1} a_j \gamma(j). \quad (11)$$

For a general lag k , the autocovariance $\gamma(k)$ follows a similar recursive relationship. The resulting system of linear equations for the autocorrelation function, $\rho(k) = \gamma(k)/\gamma(0)$, is analogous to the Yule-Walker equations:

$$\begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p-1) \end{pmatrix} = \frac{1 - (K-1)\beta}{K} \begin{pmatrix} a_0 & a_1 & \dots & a_{p-1} \\ a_1 & (a_0 + a_2) & \dots & a_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p-2} & a_{p-3} & \dots & a_1 \end{pmatrix} \begin{pmatrix} \rho(0) \\ \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p-1) \end{pmatrix} \quad (12)$$

For the case when $p = 2$, the equation for $\gamma(1)$ from Equation (11) reduces to:

$$\gamma(1) = \frac{1 - (K-1)\beta}{K} (a_0 \gamma(0) + a_1 \gamma(1)).$$

We divide both sides by $\gamma(0)$:

$$\rho(1) = \frac{1 - (K-1)\beta}{K} (a_0 + a_1 \rho(1)).$$

Provided that the term $(1 - (K-1)\beta)$ is non-zero, defining $\kappa = \frac{K}{1 - (K-1)\beta}$, rearranging the terms to solve for $\rho(1)$ gives:

$$\rho(1) = \frac{a_0}{\frac{K}{1 - (K-1)\beta} - a_1} = \frac{a_0}{\kappa - a_1},$$

This completes the proof. $\square$

This result indicates that the autocorrelation of the process can be modulated by adjusting the filter coefficients, a_0 and a_1 , and the mean correction parameter, β . The relationship shows that when $\kappa > a_1$, a larger weight on

the most recent value (i.e., a larger positive a_0) leads to a stronger positive lag-1 autocorrelation. A large positive β will make $\kappa < 0$, thus leading to a negative auto-correlation.

For the other general settings, we resort to numerical simulations to investigate the effects of key parameters on the statistical properties of the return series.

4.3 Effects of Parameters

We conduct a parametric analysis to investigate the influence of key model parameters on several stylized facts of financial returns by numerical simulations. Specifically, we focus on mean correction β , filter window weights, and probability mixture c , and their ability to reproduce desired levels of autocorrelation, skewness, excess kurtosis, and the volatility clustering effect.

4.3.1 Impact of the Mean Correction Parameter

The mean correction parameter β , as defined in Equation (4), serves two primary functions. First, it introduces a mean-reverting tendency in the return series. Second, by incorporating the magnitude of past shocks, it keeps the potential of volatility clustering. We proxy the clustering effect through the autocorrelation of absolute returns, $corr(|\Delta_t|, |\Delta_{t+k}|)$, as this effect is characterized by the tendency for large price changes (of either sign) to be followed by other large price changes.

We evaluate a three-theme market. The statistical properties of the simulated return series $\{\Delta_t\}$ for different values of β are summarized in Table 2. As anticipated, the simulated returns have a mean of zero and exhibit a total volatility greater than that of the individual components. In addition, it unveils how the parameter β adjusts return autocorrelation and absolute return autocorrelation.

As β increases from 0.0 to 0.8, the first-order autocorrelation of returns, $corr(\Delta_{t+1}, \Delta_t)$, shifts from positive (0.199) to negative (-0.096), while the first-order autocorrelation of absolute returns, $corr(|\Delta_{t+1}|, |\Delta_t|)$, rises substantially from 0.104 to 0.358. This demonstrates that volatility clustering can happen under a stronger mean correction mechanism. Furthermore, the model consistently generates positive excess kurtosis, although its ability to generate skewed return distribution is restricted under a uniform activation scheme.

Table 2: Impact of the Mean Reversion Parameter (β) on Statistical Properties.

Statistic	Mean Reversion Beta (β)			
	0.0	0.2	0.5	0.8
Moments				
Mean	0.000 (0.001)	-0.000 (0.001)	0.000 (0.000)	-0.000 (0.000)
Std. Dev.	0.014 (0.000)	0.014 (0.000)	0.015 (0.001)	0.018 (0.001)
Skewness	-0.009 (0.099)	-0.012 (0.118)	-0.006 (0.101)	-0.004 (0.084)
Kurtosis	0.173 (0.244)	0.176 (0.261)	0.163 (0.245)	0.208 (0.298)
Autocorrelation ($corr(\Delta_{t+k}, \Delta_t)$)				
Lag 1	0.199 (0.038)	0.113 (0.039)	-0.004 (0.039)	-0.096 (0.040)
Lag 2	0.172 (0.036)	0.095 (0.039)	0.003 (0.036)	-0.070 (0.044)
Lag 3	0.090 (0.034)	0.039 (0.035)	-0.001 (0.036)	-0.005 (0.039)
Absolute Autocorrelation ($corr(\Delta_{t+k} , \Delta_t)$)				
Lag 1	0.104 (0.043)	0.125 (0.044)	0.202 (0.049)	0.358 (0.058)
Lag 2	0.079 (0.040)	0.098 (0.042)	0.168 (0.044)	0.311 (0.055)
Lag 3	0.025 (0.035)	0.037 (0.038)	0.085 (0.042)	0.201 (0.056)

Note: Standard errors are reported in parentheses. The results are based on 300 simulation runs, each generating 1000 data points for a three-theme market. For this analysis, the weight of the filter window is fixed at (0.1, 0.4, 0.5), the probability mixture parameter c is set to 1.0, and the component volatility is assumed to be $\frac{0.2}{\sqrt{252}}$ based on a Gaussian mixture for daily returns.

4.3.2 Impact of Filter Window Weights

The filter window weights are designed to reflect the influence of immediate impact from the most recent market shock and the effects from previous shocks. As the analysis in Proposition 3, this model has an autocorrelation structure analogous to an autoregressive model. Large weights on recent shocks will generally bring in a large positive autocorrelation.

4.3.3 Impact of the Probability Mixture Parameter

The probability mixture parameter, c , defined in Equation (6), governs the persistence of return series through the theme activation process. A higher value of c indicates a more uniform activation scheme, while a lower value of

c increases the probability of resampling the current theme.

We present our simulated results in Table 3. As c decreases from 1.0 (purely uniform theme activation) to 0.0 (maximal self-reinforcing), we observe a pronounced increase in the persistence of both the returns and their absolute values. Specifically, the first-order autocorrelation, $\text{corr}(\Delta_{t+1}, \Delta_t)$, rises from 0.108 to 0.458, and the autocorrelation of absolute returns increases from 0.122 to 0.389. This heightened persistence also leads to fatter tails in the return distribution, as indicated by the increase in excess kurtosis from 0.169 to 0.688. Therefore, the parameter c provides a direct mechanism to control the memory and leptokurtosis of the return series.

Table 3: Impact of the Probability Mixture Parameter (c) on Statistical Properties.

Statistic	Prob Mixture c			
	0.0	0.3	0.7	1.0
Moments				
Mean	-0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	-0.000 (0.001)
Std. Dev.	0.018 (0.001)	0.016 (0.001)	0.015 (0.001)	0.014 (0.000)
Skewness	-0.012 (0.243)	-0.016 (0.188)	0.008 (0.140)	0.007 (0.101)
Kurtosis	0.688 (0.633)	0.512 (0.537)	0.261 (0.358)	0.169 (0.226)
Autocorrelation				
AC(1)	0.458 (0.048)	0.326 (0.045)	0.185 (0.043)	0.108 (0.039)
AC(2)	0.422 (0.049)	0.289 (0.042)	0.158 (0.037)	0.088 (0.037)
AC(3)	0.310 (0.054)	0.179 (0.042)	0.080 (0.039)	0.039 (0.032)
Absolute Autocorrelation				
AC(1)	0.389 (0.071)	0.280 (0.061)	0.173 (0.052)	0.122 (0.044)
AC(2)	0.341 (0.069)	0.236 (0.056)	0.140 (0.049)	0.092 (0.038)
AC(3)	0.232 (0.073)	0.137 (0.055)	0.063 (0.047)	0.036 (0.034)

Note: Standard errors are reported in parentheses. The mean reversion parameter β is held constant at 0.2. All other simulation parameters, such as the number of runs, data points per run, weighting window, and component volatility, are identical to those described in Table 2.

With different combinations of mean correction β , filter window weights $\mathbf{a}$, and probability mixture c , the model can display a wide range of behaviors,

approximating the real-world scenarios.

5 Experiments and Results

This section presents an empirical application of the model detailed in Section 4 on West Texas Intermediate (WTI) crude oil futures data. We build a four-theme market (themes are supply, demand, geopolitics and currency). We first outline the trading strategies and performance metrics, then describe the experimental design, and present and discuss the results.

5.1 Trading Strategies and Performance Metrics

We evaluate three distinct classes of trading strategies, categorized by their information source. We denote the position at time t as p_t , a lookback window of size w .

- **Group A: Passive Benchmark.** A long-only, buy-and-hold strategy that serves as a baseline to measure the intrinsic performance of the underlying asset. $p_t = 1$.
- **Group B: Price-Based Strategies.** These strategies generate trading signals exclusively from the price filtration $\mathbb{G}$. We evaluate variants of momentum-based strategy and a strategy based on ARMA modeling:
 - **Ratio Momentum:** The trading position is determined by the sign of cumulative return over a specified lookback period. $p_t = \text{sgn}(\prod_{j=1}^w (1 + \Delta_{t-j}) - 1)$
 - **Slope Momentum:** The trading position is determined by the sign of the slope of a linear regression fitted to prices over a lookback period. $p_t = \text{sgn}(\text{slope}_{t-w, t-1})$
 - **Rolling ARMA:** At each time step t , multiple ARMA(p, q) models are fitted to the data within a rolling lookback window. The model with the lowest Akaike Information Criterion (AIC) is selected to generate a one-step-ahead forecast. The trading position is the sign of this forecast.
 - **Residual Strategy:** A strategy based on the reversion of price changes to their short-term mean over a past window of width w . The position p_t is determined by the sign of the lagged residual, $p_t = \text{sgn}(\text{res}_{t-1})$. $\text{res}_{t-1} = \Delta_{t-1} - \frac{1}{w} \sum_{k=1}^w \Delta_{t-1-k}$.

- **Group C: News-Augmented Strategies.** These strategies incorporate information from the news-augmented filtration $\mathbb{F}$. We consider two variants to represent the effect of probability-mixture parameter in Equation (6).

- **Raw Signal:** The position is determined by the sign of the score of the single most dominant theme, $S_{i^*,t}$, where $i_t^* = \operatorname{argmax}_{i \in \{1, \dots, K\}} |S_{i,t}|$. Then $p_t = \operatorname{sgn}(S_{i^*,t-1})$. This corresponds to a "winner-takes-all" scenario ($c = 0$ in Equation (6)).
- **Mean Signal:** The position is determined by the sign of the average theme score, $\bar{S}_t = \frac{1}{K} \sum_{i=1}^K S_{i,t}$, $p_t = \operatorname{sgn}(\bar{S}_{t-1})$. This corresponds to a fully diversified signal ($c = 1$ in Equation (6)).

To calculate the theme score in news-augmented strategies (Group C), we choose a w -days rolling mean before time t as the consensus: $C(t) = \frac{1}{w} \sum_{i=1}^w \Delta_{t-i}$. This procedure is presented in Algorithm 1.

Algorithm 1: Theme Score Calculation

Input: Price and News Category History $\{\Delta_t, D_t\}_{t=1,2,\dots,T}$;

lookback window size w , filter window weights $\mathbf{a}$, mean correction factor β ;

Output: Theme Score $\{S_{i,t}\}_{i=1,2,\dots,K;t=1,2,\dots,T}$

```

1 for  $t = w + 1$  to  $T$  do
2   Let  $\{\Delta_j, D_j\}_{j=t-w}^{t-1}$  be the data in the lookback window;
3   Calculate consensus  $C(t)$  based on  $\{\Delta_j\}_{j=t-w}^{t-1}$ ;
4   Calculate the temporal difference  $\tilde{\Delta}_t = \Delta_t - C(t)$ ;
5   Allocate the residual  $\tilde{\Delta}_t$  across news categories based on activation
   set  $D_t$ , get theme-specific information increment  $\Delta_{i,t}$ ;
6 end
7 Apply filter window to aggregate theme-specific information
  increments series  $\{\Delta_{i,t}\}_{t=1,2,\dots,T}$  in rolling windows, collect the theme
  scores series  $\{S_{i,t}\}$  for each theme  $i$ ;
8 return Theme Score  $\{S_{i,t}\}_{i=1,2,\dots,K;t=1,2,\dots,T}$ 

```

We evaluate strategy performance using three standard metrics:

1. **Cumulative Return (R_T):** The total geometric return over the invest-

ment horizon, calculated as:

$$R_T = \prod_{t=0}^{T-1} (1 + p_{t+1} \Delta_{t+1}) - 1 \quad (13)$$

where p_{t+1} is the position held for day $t + 1$.

2. **Annualized Sharpe Ratio (SR)**: A measure of risk-adjusted return. It is calculated as:

$$SR = \sqrt{252} \times \frac{\mathbb{E}[r_t^s] - r}{\sigma(r_t^s)} \quad (14)$$

where $r_t^s = p_t \Delta_t$ is the daily strategy return, and we assume a daily risk-free rate r of $\frac{0.05}{252}$.

3. **Maximum Drawdown (MDD)**: The largest peak-to-trough decline in the portfolio's value, expressed as a percentage. It is defined as:

$$MDD = \min_{t \in [0, T]} \left(\frac{V_t - \max_{s \in [0, t]} V_s}{\max_{s \in [0, t]} V_s} \right) \quad (15)$$

where V_t is the cumulative value of the strategy at time t , represented by $V_t = 1 + R_t$.

5.2 Experimental Design

We partition our dataset into two contiguous periods for parameter tuning and validation. The initial 300 data points serve as an in-sample tuning period for hyperparameter optimization. The remaining data constitute an out-of-sample validation period, on which we evaluate the performance of the strategies using the optimized parameters.

For the price-based strategies (Group B), we optimize the lookback window hyperparameter by performing a grid search to maximize the in-sample Sharpe ratio. The optimal lookback periods were 75 days for ratio-based momentum, 5 days for slope-based momentum, and 20 days for the ARMA model.

For the news-augmented strategies (Group C), we jointly optimize the filter weights $\mathbf{W}$ (from Equation (1)), the consensus lookback window w (from Equation (3)) and the mean-correction factor β (from Equation (4)). The optimal parameter set is: $w = 5$ days, $\beta = 0.8$, and filter weights before unification are $(a_2, a_1, a_0) = (0.2, 0.2, 1.0)$. The complete results of the parameter optimization experiments are available upon request from the author.

5.3 Results and Discussion

We apply optimized hyperparameters and evaluate trading strategies’ performances. Table 4 summarizes performance across the tuning, validation, and full-sample periods. Over the full sample (2021/11/01–2025/05/30), the WTI continuous futures price declined by 33%. Against this adverse backdrop, the two news-enhanced signals (Raw and Mean Signal; Group C) and the two price-only momentum benchmarks (Slope- and Ratio-based; Group B) delivered positive performance. Among them, the Raw Signal strategy exhibits the strongest overall performance, attaining the highest cumulative return and Sharpe ratio in both the validation period and the full sample, while also limiting maximum drawdowns relative to buy-and-hold. These results also indicate that theme-based news signals contain incremental, tradable information beyond standard momentum. In contrast, the residual-based specification is unstable, and the ARMA-based signal fails to extract reliable predictability from crude oil returns, producing weak risk-adjusted performance.

The optimal hyperparameters unveil aligned market schemes. The best-performing Raw Signal and Slope Momentum configurations employ a five-day lookback and filter weights (0.2, 0.2, 1.0), pointing to myopic attention and rapid decay of news relevance. The optimal mean-correction parameter $\beta = 0.8$ in the combination step implies strong mean reversion in attention diffusion toward inactive themes. This magnitude accords with volatility clustering and with the negative second-order autocorrelation in real returns, which reflects short-term reversal dynamics.

Figure 4 plots cumulative returns during validation period. Following a peak in September 2023, WTI prices entered a pronounced downtrend. The news-augmented strategies (Raw Signal in blue; Mean Signal in red) generated most of their gains from May–December 2023 and again after May 2025, while experiencing drawdowns during 2024. Over that interval, the Slope Momentum strategy captured the prevailing trend more effectively.

5.4 Performance Enhancement via a Theme-Ranked Regression Model

Winner-takes-all setting (raw-signal) and uniform activation setting(mean-signal) both achieve positive returns. In this section, we evaluate if a smart combination of both methods could further improve the momentum strategy. We construct the weights by regressing next-day returns on the cross-section of daily theme scores, ranked by absolute magnitude. The working hypothesis

Table 4: Performance Metrics of Trading Strategies

Panel A: Parameter Tuning Period 2021/11/01 - 2022/12/14 (280 days)							
Metric	Raw Signal	Mean Signal	Residual Signal	Long Only	Ratio Mom.	Slope Mom.	ARMA Signal
Cumulative Returns	0.35	1.08	0.77	-0.14	0.47	0.31	0.02
Sharpe Ratio	0.70	1.50	1.20	-0.13	0.89	0.64	-0.05
Maximum Drawdown	-0.38	-0.23	-0.21	-0.46	-0.24	-0.29	-0.26
Panel B: Validation Period 2022/12/27 - 2025/05/30 (604 days)							
Cumulative Returns	1.08	0.14	-0.36	-0.25	-0.02	0.32	-0.33
Sharpe Ratio	0.97	0.17	-0.59	-0.38	-0.03	0.37	-1.18
Maximum Drawdown	-0.30	-0.38	-0.53	-0.41	-0.36	-0.28	-0.38
Panel C: Full Period 2021/11/01 - 2025/05/30 (891 days)							
Cumulative Returns	1.90	1.30	0.13	-0.33	0.39	0.78	-0.31
Sharpe Ratio	0.85	0.68	0.15	-0.25	0.30	0.49	-0.76
Maximum Drawdown	-0.38	-0.51	-0.53	-0.58	-0.36	-0.29	-0.48

Note: This table presents performance metrics for seven trading strategies. Raw Signal and Mean Signal represent news-enhanced strategies (Group C), using the sign of the most dominant score and the sign of the mean score across themes respectively. Residual (Group B) uses the deviation from rolling mean as the trading signal. Long Only is the passive buy-and-hold strategy. Ratio Mom. and Slope Mom. (Group B) are momentum strategies based on, respectively, the ratio of cumulative returns at the window endpoints and the slope of daily returns within a rolling window. ARMA Signal is derived from rolling ARMA forecasts. Cumulative Returns represent the total geometric return. The Sharpe Ratio is annualized. The best performance for each metric in each panel is highlighted in bold.

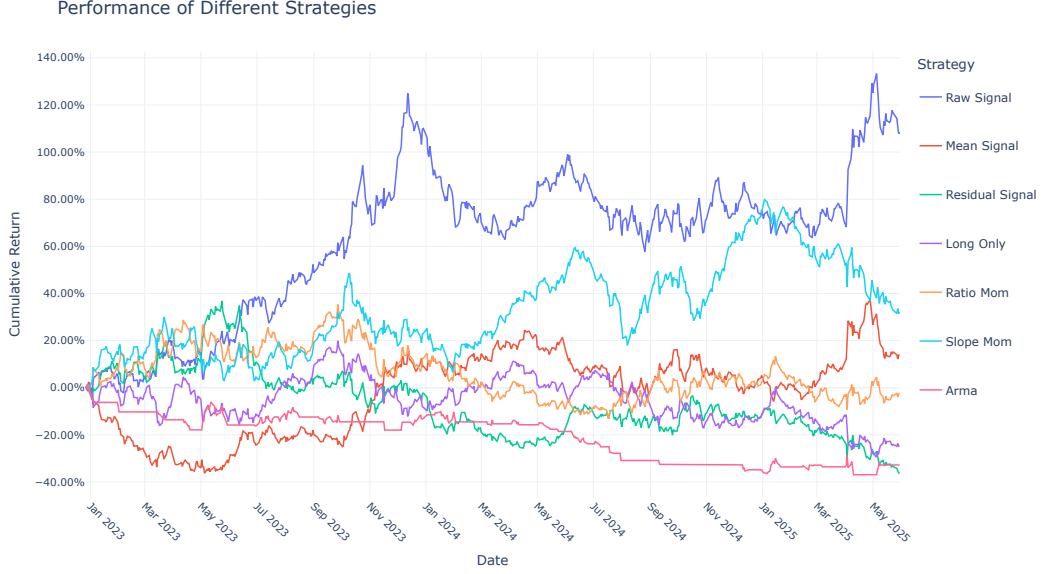


Figure 4: Trading Performance during the validation period. This figure reports the cumulative returns of the strategies during the validation period (2022/12/27 - 2025/05/30).

is that themes with larger absolute scores have greater impact for subsequent returns.

For each day t , let $(S_{i,t})_{i=1}^K$ denote the theme scores. We construct ranked features $(\tilde{S}_{i,t})_{i=1}^K$ by ordering $\{|S_{i,t}|\}$ in descending order: $\tilde{S}_{1,t}$ is the score with the largest absolute value, $\tilde{S}_{2,t}$ the second largest, and so forth, with signs preserved. Instead taking the largest absolute value or average theme score to determine the sign of trading position, our decision rule is based on a linear combination of all theme scores. Specifically, we regress the next-day return, Δ_{t+1} on these ranked scores:

$$\Delta_{t+1} = \alpha + \sum_{i=1}^K \beta_i \tilde{S}_{i,t} + \epsilon_{t+1}. \quad (16)$$

We expect larger coefficients for top-ranked features (i.e., smaller i). The fitted model yields a refined trading signal:

$$p_{t+1} = \text{sgn} \left(\hat{\alpha} + \sum_{i=1}^K \hat{\beta}_i \tilde{S}_{i,t} \right), \quad (17)$$

where p_{t+1} is the position at time $t+1$, and $\hat{\alpha}$ and $\hat{\beta}_i$ are estimated parameters.

We first estimate the model, Equation (16), over the full sample. A declin-

ing pattern across the estimated coefficients of the reindexed features in Table 5 is consistent with the hypothesis. To avoid look-ahead bias and evaluate out-of-sample performance, we re-estimate the model on the first 400 observations. Using the training-period coefficients in Equation (17), we generate positions over the test window (May 2023-May 2025). The refined strategy achieves a cumulative return of 91% versus 74% for the benchmark, a higher Sharpe ratio (1.05 vs. 0.90), and a smaller maximum drawdown (-0.25 vs. -0.29), with trajectories displayed in Figure 5.

Table 5: OLS Regression Results on the whole period (2021/10 to 2025/05).

	Coefficient	Std. Error
Constant	0.0000	0.001
Ranked Score 1	0.0529	0.056
Ranked Score 2	0.0946**	0.046
Ranked Score 3	0.0003	0.077
Ranked Score 4	-0.1246*	0.076
Observations	904	
R-squared	0.012	

Note: Ranked Score i corresponds to i -th ranked feature $\tilde{S}_{t,i}$. Standard errors are HAC (Newey–West) with 3 lags. Significance level are denoted as: *($p \leq 0.10$), and **($p \leq 0.05$).

Furthermore, to collect statistical evidence for discrepancies in predictive accuracy across four forecasting rules, we apply the Diebold–Mariano (DM) test (1995) to: (i) theme-ranked regression, (ii) Raw Signal, (iii) slope-based momentum, and (iv) ratio-based momentum.

Given two forecasts $\hat{y}_{1t}$ and $\hat{y}_{2t}$ for period t and the realized outcome y_t , define forecast errors under a 0–1 loss as

$$e_{it} = 1 - \mathbb{I}\{y_t = \hat{y}_{it}\}, \quad i \in \{1, 2\},$$

where $\hat{y}_{it} \in \{-1, 1\}$ denotes the predicted position at time t , and $y_t = \text{sgn}(\Delta_t)$ is the sign of the realized return. The loss differential is $d_{12,t} = e_{1t} - e_{2t}$ with null hypothesis assuming equal expected predictive accuracy.

$$H_0 : \mathbb{E}[d_{12,t}] = 0.$$

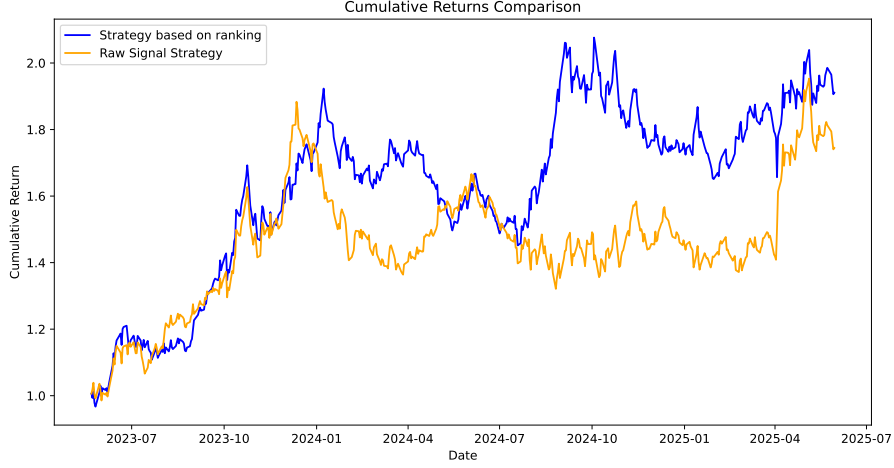


Figure 5: Cumulative Performance Comparison. This figure shows the cumulative performance of the refined regression-based strategy (blue) and the raw-signal strategy (yellow) over the test period (2023/05/22–2025/05/30).

Let $\bar{d}_{12} = T^{-1} \sum_{t=1}^T d_{12,t}$ and $\hat{\sigma}_{\bar{d}_{12}}$ be the HAC (Newey–West) standard error of $\bar{d}_{12}$ with lag length $T^{1/3}$. The DM statistic,

$$DM_{12} = \frac{\bar{d}_{12}}{\hat{\sigma}_{\bar{d}_{12}}},$$

is asymptotically standard normal under H_0 (Diebold and Mariano (1995)). A negative DM_{12} indicates that forecast 1 incurs a lower expected loss than forecast 2, thus forecast 1 is more accurate in predicting the sign.

We report one-sided p-values for the alternative $H_1 : \mathbb{E}[d_{12,t}] < 0$ and the DM statistics of pairwise comparisons across all strategies in Table 6. The results indicate the following ordering of forecasting performance, from best to worst: Regression-Enhanced forecast, Raw-Signal forecast, Slope-Based Momentum forecast, and Ratio-Based Momentum forecast. Regression-enhanced forecasts and Raw-Signal forecasts show statistically significant improvements over the ratio-based benchmark at 5% and 10% levels, respectively.

6 Conclusion

Whether news narratives provide incremental information beyond that contained within historical return series and, if so, how this information can be effectively extracted to enhance return prediction? Our paper provides a perspective by proposing a theme-based return decomposition model.

In our model, we leverage price drivers from news to label return data,

Table 6: Pairwise Diebold–Mariano (DM) tests across strategies.

	Raw Signal	Slope Mom.	Ratio Mom.
Regression	-0.253	-1.042	-1.696**
Raw Signal		-0.830	-1.380*
Slope Mom.			-0.532

Notes: Entry (row i , column j) is the DM statistic comparing strategy i against strategy j ; negative values favor the row strategy. Significance levels are denoted as: $*p \leq 0.10$, $**p \leq 0.05$. One-sided p-values for the “row better than column” alternative. Standard errors are HAC (Newey–West) with lag $T^{1/3}$. Regression is the regression-enhanced forecast, Raw Signal is the forecast based on dominant theme score, Ratio Mom. is the ratio-based momentum forecast and Slope Mom. is the slope-based momentum forecast.

thereby enabling a granular decomposition of price movements. We propose a mixture model for return generation that explicitly links price dynamics to underlying, theme-specific market attention. Empirical validation on crude oil futures data demonstrates the effectiveness and robustness of our model, which significantly outperforms traditional momentum strategies based solely on historical return time series.

In addition, the model’s fitted parameters reveal aligned characteristics of crude oil return dynamics. In our estimation, the market exhibits short-term attention parameterized by a quickly waning filter window weights and a mean-reversion nature parameterized by a high mean-correction β parameter.

Our return decomposition scheme links price movement with its underlying drivers and our theme score based asset allocation diagram provides an effective approximation of market scheme.

We present our data at <https://juwyang.github.io/UMCO/>, tracking news theme decomposition for the WTI crude oil market, together with identified price drivers, market themes, and market theme scores. The code is available at <https://github.com/juwyang/UMCO>.

References

- W. Antweiler and M. Z. Frank. Is all that talk just noise? the information content of internet stock message boards. *The Journal of finance*, 59(3):1259–1294, 2004.
- S. R. Baker, N. Bloom, and S. J. Davis. Measuring economic policy uncertainty. *The quarterly journal of economics*, 131(4):1593–1636, 2016.
- H. Bessembinder, J. F. Coughenour, P. J. Seguin, and M. M. Smoller. Mean reversion in equilibrium asset prices: Evidence from the futures term structure. *The Journal of Finance*, 50(1):361–375, 1995.
- J. Bollen, H. Mao, and X.-J. Zeng. Twitter mood predicts the stock market. *Journal of computational science*, 2(1):1–8, 2011.
- L. Bybee, B. Kelly, A. Manela, and D. Xiu. Business news and business cycles. *The Journal of Finance*, 79(5):3105–3147, 2024.
- K. Daniel, D. Hirshleifer, and A. Subrahmanyam. Investor psychology and security market under-and overreactions. *the Journal of Finance*, 53(6):1839–1885, 1998.
- S. Delle Chiaie, L. Ferrara, and D. Giannone. Common factors of commodity prices. *Journal of Applied Econometrics*, 37(3):461–476, 2022.
- F. X. Diebold and R. S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 1995.
- G. Doyle and C. Elkan. Financial topic models. In *Working notes of the NIPS-2009 workshop on applications for topic models: Text and beyond workshop*, volume 210, 2009.
- K. Du, Y. Zhao, R. Mao, F. Xing, and E. Cambria. Natural language processing in finance: A survey. *Information Fusion*, 115:102755, 2025.
- E. F. Fama and K. R. French. Commodity futures prices: Some evidence on forecast power, premiums, and the theory of storage. *The Journal of Business*, 60(1):55–73, 1987.
- M. Gentzkow, B. Kelly, and M. Taddy. Text as data. *Journal of Economic Literature*, 57(3):535–574, 2019.

- G. Gorton and K. G. Rouwenhorst. Facts and fantasies about commodity futures. *Financial Analysts Journal*, 62(2):47–68, 2006.
- A. Goyal, N. Jegadeesh, and A. Subrahmanyam. Empirical determinants of momentum: A perspective using international data. *Review of Finance*, 29(1):241–273, 2025.
- H. Hong and J. C. Stein. A unified theory of underreaction, momentum trading, and overreaction in asset markets. *The Journal of finance*, 54(6):2143–2184, 1999.
- N. Jegadeesh and D. Wu. Word power: A new approach for content analysis. *Journal of financial economics*, 110(3):712–729, 2013.
- Z. T. Ke, B. T. Kelly, and D. Xiu. Predicting returns with text data. Technical report, National Bureau of Economic Research, 2019.
- F. Li. Annual report readability, current earnings, and earnings persistence. *Journal of accounting and economics*, 45(2-3):221–247, 2008.
- T. Loughran and B. McDonald. Textual analysis in accounting and finance: A survey. *Journal of accounting research*, 54(4):1187–1230, 2016.
- A. Manela and A. Moreira. News implied volatility and disaster concerns. *Journal of Financial Economics*, 123(1):137–162, 2017.
- J. Miffre and G. Rallis. Momentum strategies in commodity futures markets. *Journal of Banking & Finance*, 31(6):1863–1886, 2007.
- T. J. Moskowitz, Y. H. Ooi, and L. H. Pedersen. Time series momentum. *Journal of financial economics*, 104(2):228–250, 2012.
- R. J. Shiller. Narrative economics. *American Economic Review*, 107(4), 2017.
- K. Somefun, R. Perchet, C. Yin, and R. Leote de Carvalho. Allocating to thematic investments. *Financial Analysts Journal*, 79(1):18–36, 2023.
- P. C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *The Journal of finance*, 62(3):1139–1168, 2007.
- F. Z. Xing, E. Cambria, and R. E. Welsch. Natural language based financial forecasting: a survey. *Artificial Intelligence Review*, 50(1):49–73, 2018.

7 Appendix

7.1 Descriptive Statistics of Thematic Returns

We build synthetic theme-specific return series, and record their descriptive statistics in Table 7. Let $\{\Delta_t\}$ be the time series of daily returns and $\mathcal{T}$ be the set of themes under consideration. For each time t , a subset of themes $D_t \subseteq \mathcal{T}$ is considered active. We decompose the daily return into theme-specific series $\{X_{i,t}\}$ for each theme $i \in \mathcal{T}$ as follows:

$$X_{i,t} = \Delta_t \cdot \mathbb{I}\{i \in D_t\},$$

where $\mathbb{I}(\cdot)$ is the indicator function. For each theme, we compute the moments only over its activated periods from a condensed thematic time series: $\{X_{i,t} | i \in D_t\}$. The total sample size from 2021-10-01 to 2025-05-30 is 911. After the theme decomposition, we noticed geopolitics and supply related time have positive means while demand-related time has a negative mean during this period. For autocorrelation, geopolitics theme and demand theme display strong trend-continuation patterns, with positive first-order autocorrelation, while currency-related time has a negative first-order autocorrelation. The gross return has displayed positive first-order autocorrelation ($AC(1) = 0.024$) and a negative second-order autocorrelation ($AC(2) = -0.061$) of comparable magnitude.

Table 7: Descriptive Statistics of Daily and Thematic Returns.

	Return	Currency	Demand	Supply	Geopolitics
Count	911	297	417	311	206
Mean (in %)	-0.006	0.001	-0.544	0.429	0.512
SD (Ann.)	0.374	0.323	0.371	0.343	0.411
Skew	-0.494	-0.307	-0.703	-0.153	0.040
Kurt	2.189	0.733	1.774	1.169	0.940
AC(1)	0.024	-0.087	0.074	0.020	0.099
AC(2)	-0.061	-0.099	0.053	0.021	0.110

Note: This table reports descriptive statistics for the full sample of daily returns and for the theme-specific return series. The moments for each theme are calculated on a condensed series, using only observations from days when the theme was active. Count indicates the number of active days. SD (Ann.) is the annualized standard deviation. Skew is skewness, Kurt is excess kurtosis calculated by subtracting 3 from the kurtosis value, and AC(1) and AC(2) are the first and second-order autocorrelations. The sample period is October 2021 to May 2025.